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Demonstrating the Ratio of Time Dilation Between Two Systems Emerges Solely From Their Relative Respective Mass to Radius Ratios

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ABSTRACT: General relativity tells us that time is relativistic — it only has meaning when compared to another system. However, in the standard interpretation of general relativity, it feels very much like one can simply calculate the relativistic time dilation for a single system. That is in fact a misconception; if you look carefully at the Schwarzschild threshold condition for gravitational collapse, it is actually comparing two systems: one, the stated system being evaluated, and two, that same system at its Schwarzschild radius according to its mass. A gravitational time dilation figure is returned; that value is mathematically identical to the ratio of the gravitational time dilation of one system to the gravitational time dilation of the other. We demonstrate and validate this principle for the above example in addition to four other distinct cases that span the classical to relativistic regimes: comparing two systems of constant radius, comparing two systems of constant inertia, comparing two systems of constant density, and comparing two systems of distinct mass and radius.

Keywords: time dilation, general relativity, special relativity, gravitational time dilation, Lorentz time dilation, DeGerlia compactness, inertial density, moment of inertia, Schwarzschild radius

1 Introduction

The Schwarzschild gravitational time-dilation calculation $\sqrt{1-r_s/R}$ contains two radii: R, the radius of a system, and $r_s = 2GM/c^2$, the Schwarzschild radius of the same mass. Same mass at two different radii: the formula compares two systems. The notation places r_s inside one symbol, which gives the formula a single-system appearance, but the calculation is two-system. The single-system reading gives the right number — the formula carries both systems — and is incoherent within relativity, where a frame of reference requires a comparator.

This paper writes both systems openly. For each uniform spherical system, the DeGerlia compactness $D = M/R$ (kg/m) is the mass-to-radius ratio[1]. The DeGerlia compactness ratio $k_d = D_1/D_2 = M_1R_2/(M_2R_1)$ governs the time dilation between two such systems. The static-mass case ($M_1 = M_2$, different radii) is the standard gravitational time-dilation calculation[2,3]. The static-radius case ($R_1 = R_2$, different masses) is the standard Lorentz time-dilation calculation[4]. Six worked cases in Section 6 and the appendix tables (Appendix Appendix A) demonstrate that calculating through the standard Schwarzschild form $\sqrt{1-r_s/R}$ and through the $M_1R_2/(M_2R_1)$ ratio yields the same result across the classical and relativistic regimes. Agreement is not asserted on trust: every input and output is tabulated in the appendix, and each row is reproducible directly from $r_s = 2GM/c^2$ and the standard Schwarzschild form.

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2 Inertial Relativity

A system, in this context, is any collection of mass and spatial extent one chooses to define.

2.1 Rotational Inertia

Moment of inertia dictates how an applied torque translates into angular acceleration. It is defined as:

$$I = \sum_i m_i r_i^2 \tag{1}$$

where r_i is the perpendicular distance from each mass element m_i to the rotation axis. Just as mass governs how a system responds to linear force ($F = ma$), moment of inertia governs how it responds to torque ($\tau = I\alpha$).

2.2 Linear Inertia

Linear inertia is mass:

$$M = \sum_i m_i \tag{2}$$

where m_i are the constituent mass elements. Linear inertia defines how an applied force translates into translational acceleration. It is quantified entirely by mass ($F = ma$). It is dimensional, operating along a specific spatial vector, but unlike moment of inertia, it is isotropic: mass offers the same resistance to changes in motion regardless of the direction of the vector.

2.3 Isometric Scaling

Isometric scaling is the uniform scaling of a system in all spatial dimensions while maintaining constant density. Under isometric scaling, every length in the system scales by the same factor k:

$$\ell' = k\ell \tag{3}$$

where ℓ is the characteristic length of the system and k is the linear scale factor. Isometric scaling of a system by a factor of k results in the following changes to mass, radius, and moment of inertia:

$$M' = k^3 M \tag{4}$$

where M is the mass and k is the linear scale factor.

$$R' = kR \tag{5}$$

where R is the radius.

$$I' = k^5 I \tag{6}$$

where I is the moment of inertia and k is the linear scale factor.

2.4 Inertia Abstracts Mass and Radius

Because every change in length necessarily changes the system's mass, Equations 3–5 hold only for strict isometric scaling; Equation 6, however, abstracts mass and radius into moment of inertia. If two systems share moment of inertia, then an identical torque applied to each system will produce identical changes in motion, regardless of the system geometry; independent of mass and radius individually.

3 Inertial Density P

Inertial density P[1] is the moment of inertia of a system divided by its volume:

$$P = I/V \tag{7}$$

Since linear inertia is mass, this reduces directly to mass density $\rho = M/V$ in the linear case; inertial density is the rotational generalization of mass density, a quantity already familiar from linear mechanics. Or more accurately, mass density is simply the linear interpretation of inertial density.

Substituting $I = \frac{2}{5}MR^2$ and $V = \frac{4}{3}\pi R^3$, with R the radius, this reduces to:

$$P = \frac{M}{R} \times \frac{3}{10\pi} \tag{8}$$

Just as ordinary density ($\rho = M/V$) is mass distributed over volume, inertial density is rotational inertia distributed over volume. As stated in the prior work: *mass and density are to linear motion as moment of inertia and inertial density are to rotational motion.*

3.1 DeGerlia Compactness D

$D = M/R$ is a simplified analog to inertial density, related to P by the geometric factor $3/10\pi$, and represents the mass over radius ratio for a system. Given that this quantity is almost always used in a ratio that would cancel out all geometric vectors and constants, for ease of use, this metric D for inertial density is typically used over the geometrically accurate inertial density, P.

3.2 DeGerlia Threshold D_{crit}

The Schwarzschild condition can be expressed as a static universal constant threshold of DeGerlia compactness D. There are several advantages to this approach, not the least of which is that you don't have to calculate the threshold for every mass. And, it's much easier to calculate D than P. Inertial density is an intuitive metric that's easier to work with.

Setting $R = r_s$ (the Schwarzschild condition)[3] and solving yields the **DeGerlia Threshold**: the universal constant of DeGerlia compactness at which the escape velocity equals the speed of light:

$$D_{crit} = \frac{c^2}{2G} = (6.73295 \pm 0.00015) \times 10^{26}\ \text{kg/m} \tag{9}$$

Note this value is exact and accurate to the limit of precision imposed by G, which is accurate to six significant figures with a margin of error of $15/10^6$[5].

4 The Space-Time Equivalence

The following identity relates the time-dilation ratio of two uniform spherical systems to their mass and radius.*

$$k_d = \frac{I_1}{I_2} \cdot \left(\frac{R_2}{R_1}\right)^3 = \frac{D_1}{D_2} = \frac{M_1 R_2}{M_2 R_1} \tag{10}$$

where I_1, I_2 are the moments of inertia, R_1, R_2 are the radii, M_1, M_2 are the masses, and $D_i = M_i/R_i$ is the DeGerlia Compactness of system i. The three forms in Equation 10 are algebraically identical; $(I_1/I_2)(R_2/R_1)^3$ exposes the inertia-radius structure, D_1/D_2 exposes the compactness-ratio structure, and $M_1R_2/(M_2R_1)$ is the fully expanded form.

The pace of time has meaning only relative to another pace of time. The DeGerlia compactness ratio k_d governs the relative time-dilation between two uniform spherical systems. The standard gravitational time-dilation calculation takes the Schwarzschild form $\sqrt{1-k_s}$, where $k_s = D/D_{crit} = r_s/R$. The DeGerlia compactness ratio $k_d = D_1/D_2$ and the Schwarzschild form k_s are the same scale factor k evaluated by two routes; the subscript denotes which route produced it.

* This identity has been previously referred to as the "space-time equivalence"[6].

Via the Schwarzschild radius, the same scale factor is:

$$k_s = \frac{D}{D_{crit}} = \frac{r_s}{R} \tag{11}$$

where $D = M/R$ is the system's DeGerlia Compactness and $r_s = 2GM/c^2$ is its Schwarzschild radius. k_s is the Schwarzschild-route name for k that appears in the standard $\sqrt{1-k}$ for GTD.

4.1 Relating Linear Scale Factor k to General Relativity

Time dilation is always:

$$\frac{d\tau}{dt} = \sqrt{1-k} \tag{12}$$

where $d\tau/dt$ is the time-dilation factor between the two systems and k is the DeGerlia compactness ratio for the case being computed. The specific forms are:

- $k_d = D_1/D_2$ (Equation 10) — via inertial density ratio
- $k_s = D/D_{crit} = r_s/R$ (Equation 11) — via Schwarzschild radius

Same k symbol across forms: a given numerical k produces the same time-scale-factor result regardless of which form it appears in.

4.2 STE Example

Setup. A uniform sphere of mass 1 kg and radius 1 m (System 2) is scaled isometrically by a factor of two, producing a sphere of mass $2^3 = 8\,\text{kg}$ and radius 2 m (System 1). The linear scale factor between them is $k = R_1/R_2 = 2$ by construction.

Change in inertia. The moments of inertia are

$$I_1 = \tfrac{2}{5}(8)(2)^2 = 12.8\,\text{kg}\,\text{m}^2, \qquad I_2 = \tfrac{2}{5}(1)(1)^2 = 0.4\,\text{kg}\,\text{m}^2,$$

so $I_1/I_2 = 32 = 2^5 = k^5$ — the moment of inertia changes by the fifth power of the linear scale factor.

Gravitational time dilation. Each system's Schwarzschild radius is $r_s = 2GM/c^2$, giving $r_{s,1} = 1.18819 \times 10^{-26}$ m and $r_{s,2} = 1.48523 \times 10^{-27}$ m. The Schwarzschild gravitational time dilation $gtd = \sqrt{1 - r_s/R}$ is then

$$gtd_1 = \sqrt{1 - \frac{r_{s,1}}{R_1}} = 1 - 2.97046 \times 10^{-27},$$

$$gtd_2 = \sqrt{1 - \frac{r_{s,2}}{R_2}} = 1 - 7.42616 \times 10^{-28},$$

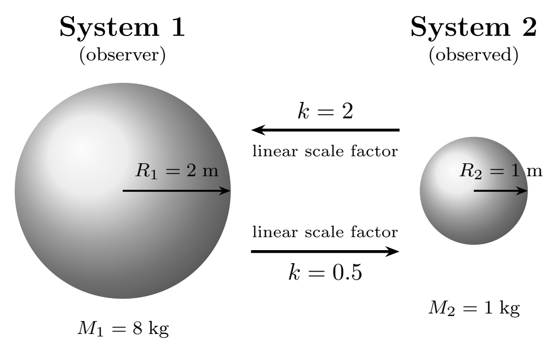
so the relative time dilation between the two systems is

$$\frac{gtd_1}{gtd_2} = 1 - 2.22785 \times 10^{-27}.$$

DeGerlia compactness ratio. The ratio k_d follows from the DeGerlia compactness $D = M/R$ ($D_1 = 4\,\text{kg/m}$, $D_2 = 1\,\text{kg/m}$):

$$k_d = \frac{D_1}{D_2} = \frac{4}{1} = 4 = k^2, \qquad \sqrt{k_d} = 2 = k.$$

This matches the STE Example column of Appendix Appendix A: the gravitational time dilation ratio is $1 - 2.22785 \times 10^{-27}$ and $\sqrt{k_d} = 2$.



Two uniform spheres related by isometric scaling, scale factor $k = 2$: System 1 ($M_1 = 8\,\text{kg}$, $R_1 = 2\,\text{m}$) and System 2 ($M_2 = 1\,\text{kg}$, $R_2 = 1\,\text{m}$). The moment of inertia changes by $2^5 = 32$; the relative time dilation $1 - 2.22785 \times 10^{-27}$ is obtained equivalently from the Schwarzschild and inertial-density formulations.

5 General and Special Relativity

Schwarzschild gravitational time dilation:

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{r_s}{R}} \tag{13}$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius and R is the radius.

Lorentz time dilation:

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{v^2}{c^2}} \tag{14}$$

where v is the velocity of the moving frame and c is the speed of light.

These are the same form as Equation 12, where $k = r_s/R$ in the gravitational case and $k = v^2/c^2$ in the velocity case. The two expressions are equivalent: the escape velocity at radius R is $v^2 = 2GM/R^7$, so $v^2/c^2 = 2GM/(Rc^2) = r_s/R$.

Schwarzschild gravitational time dilation is the STE (Equation 10) evaluated under the constraint $M_1 = M_2$ (same mass, different radii). Lorentz time dilation is the STE evaluated under the constraint $R_1 = R_2$ (same radius, different masses). Both are cases of the same DeGerlia-compactness-based relation; the derivations of each are developed in Section 6.

6 Expressions of Relative Inertial Linear Scale

The universal scale factor between any two systems is the DeGerlia compactness ratio k_d (Equation 10), with $k_d = D_1/D_2 = M_1R_2/(M_2R_1)$. Each subsection below applies a specific physical constraint that simplifies k_d, derives the resulting form, and demonstrates the calculation with a worked numerical example.

Numerical values are reported at six significant figures in e-style notation. Reference constants used throughout: $D_{crit} = 6.73295 \times 10^{26}$ kg/m, $G = 6.67430 \times 10^{-11}$ m^3 kg^{-1} s^{-2}, $c = 2.99792 \times 10^8$ m/s. Example parameter sets match the columns of the pair comparison tables (Appendix Appendix A).

6.1 Static Mass (Gravitational) ($M_1 = M_2$)

When the two systems share the same mass, the mass terms cancel in k_d:

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \frac{R_2}{R_1}$$

The compactness ratio reduces to a ratio of radii. Equivalently in terms of DeGerlia compactness $D = M/R$, this is $D_1/D_2 = R_2/R_1$; each system's normalized DeGerlia compactness against the DeGerlia Threshold yields $D/D_{crit} = r_s/R$, recovering the Schwarzschild gravitational time dilation parameter.

This case is the standard gravitational time-dilation calculation; the $M_1R_2/(M_2R_1)$ ratio reduces to the familiar Schwarzschild factor here. Both systems are written out and computed independently below — the same Schwarzschild calculation, with the second system (the same mass at its Schwarzschild radius) made visible instead of folded into r_s.

Example (Static Mass): S_1: $M_1 = 2.00000 \times 10^{30}$ kg, $R_1 = 3.00000 \times 10^3$ m (near the Schwarzschild radius). S_2: $M_2 = 2.00000 \times 10^{30}$ kg, $R_2 = 7.00000 \times 10^8$ m.

DeGerlia compactness ratio:

$$k_d = \frac{R_2}{R_1} = \frac{7.00000 \times 10^8}{3.00000 \times 10^3} = 2.33333 \times 10^5$$

GTD calculation: Schwarzschild gravitational time dilation directly. With $M_1 = M_2$, both systems share the same Schwarzschild radius:

$$r_{s,1} = r_{s,2} = \frac{2GM}{c^2} = \frac{2(6.67430 \times 10^{-11})(2.00000 \times 10^{30})}{(2.99792 \times 10^8)^2}$$

$$= 2.97046 \times 10^3 \text{ m}$$

GTD for each system via $\sqrt{1 - r_s/R}$:

$$\text{GTD}_1 = \sqrt{1 - \frac{r_{s,1}}{R_1}} = \sqrt{1 - \frac{2.97046 \times 10^3}{3.00000 \times 10^3}}$$

$$= \sqrt{1 - 9.90155 \times 10^{-1}} = 1 - 9.00777 \times 10^{-1}$$

$$\text{GTD}_2 = \sqrt{1 - \frac{r_{s,2}}{R_2}} = \sqrt{1 - \frac{2.97046 \times 10^3}{7.00000 \times 10^8}}$$

$$= \sqrt{1 - 4.24352 \times 10^{-6}} = 1 - 2.12176 \times 10^{-6}$$

giving the relative pace of time:

$$\frac{\text{GTD}_1}{\text{GTD}_2} = \frac{1 - 9.00777 \times 10^{-1}}{1 - 2.12176 \times 10^{-6}} = 1 - 9.00776 \times 10^{-1}$$

See Table 2, column $M = M$.

6.2 Static Radius (Linear) ($R_1 = R_2$)

When the two systems share the same radius, the radius terms cancel in k_d:

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \frac{M_1}{M_2}$$

The DeGerlia compactness ratio reduces to a ratio of masses. The Lorentz factor (Equation 14) has the same form as Equation 12, with $k = v^2/c^2$. Since velocity is distance per unit time, v^2/c^2 is the square of a ratio of two lengths, with the shared time unit cancelling. Gravitational time dilation confirms this directly: $v^2/c^2 = r_s/R$ via the escape velocity identity, so the velocity-based k of special relativity and the radius-based k of general relativity are the same length ratio.

Example (Static Radius): S_1: $M_1 = 1.00000 \times 10^{30}$ kg, S_2: $M_2 = 1.00000 \times 10^{24}$ kg, both at $R = 1.00000 \times 10^8$ m.

DeGerlia compactness ratio:

$$k_d = \frac{M_1}{M_2} = \frac{1.00000 \times 10^{30}}{1.00000 \times 10^{24}} = 1.00000 \times 10^6$$

GTD calculation: Schwarzschild gravitational time dilation directly. Each system's Schwarzschild radius is $r_s = 2GM/c^2$:

$$r_{s,1} = \frac{2GM_1}{c^2} = \frac{2(6.67430 \times 10^{-11})(1.00000 \times 10^{30})}{(2.99792 \times 10^8)^2}$$

$$= 1.48523 \times 10^3 \text{ m}$$

$$r_{s,2} = \frac{2GM_2}{c^2} = \frac{2(6.67430 \times 10^{-11})(1.00000 \times 10^{24})}{(2.99792 \times 10^8)^2}$$

$$= 1.48523 \times 10^{-3} \text{ m}$$

GTD for each system via $\sqrt{1 - r_s/R}$:

$$\text{GTD}_1 = \sqrt{1 - \frac{r_{s,1}}{R}} = \sqrt{1 - \frac{1.48523 \times 10^3}{1.00000 \times 10^8}}$$

$$= \sqrt{1 - 1.48523 \times 10^{-5}} = 1 - 7.42619 \times 10^{-6}$$

$$\text{GTD}_2 = \sqrt{1 - \frac{r_{s,2}}{R}} = \sqrt{1 - \frac{1.48523 \times 10^{-3}}{1.00000 \times 10^8}}$$

$$= \sqrt{1 - 1.48523 \times 10^{-11}} = 1 - 7.42616 \times 10^{-12}$$

giving the relative pace of time:

$$\frac{\text{GTD}_1}{\text{GTD}_2} = \frac{1 - 7.42619 \times 10^{-6}}{1 - 7.42616 \times 10^{-12}} = 1 - 7.42618 \times 10^{-6}$$

See Table 2, column $R = R$.

6.3 Constant Density (Isometric) ($\rho_1 = \rho_2$)

When the two systems share the same density, $M \propto R^3$, so $M_1/M_2 = (R_1/R_2)^3$. Substituting into k_d:

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \left(\frac{R_1}{R_2}\right)^3 \cdot \frac{R_2}{R_1} = \left(\frac{R_1}{R_2}\right)^2$$

The DeGerlia compactness ratio reduces to the square of the radius ratio. The linear scale factor follows as $\sqrt{k_d} = R_1/R_2$, recovering the isometric equivalence.

Example (Constant Density): S_1: $M_1 = 1.00000 \times 10^{30}$ kg, $R_1 = 1.00000 \times 10^8$ m. S_2: $M_2 = 1.00000 \times 10^{27}$ kg, $R_2 = 1.00000 \times 10^7$ m. Both at $\rho = 2.38732 \times 10^5$ kg/m^3.

Compactness ratio and linear scale factor:

$$k_d = \left(\frac{R_1}{R_2}\right)^2 = 1.00000 \times 10^2, \qquad \sqrt{k_d} = \frac{R_1}{R_2} = 1.00000 \times 10^1$$

GTD calculation: Schwarzschild gravitational time dilation directly. Each system's Schwarzschild radius is $r_s = 2GM/c^2$:

$$r_{s,1} = \frac{2GM_1}{c^2} = \frac{2(6.67430 \times 10^{-11})(1.00000 \times 10^{30})}{(2.99792 \times 10^8)^2}$$

$$= 1.48523 \times 10^3 \text{ m}$$

$$r_{s,2} = \frac{2GM_2}{c^2} = \frac{2(6.67430 \times 10^{-11})(1.00000 \times 10^{27})}{(2.99792 \times 10^8)^2}$$

$$= 1.48523 \times 10^0 \text{ m}$$

GTD for each system via $\sqrt{1 - r_s/R}$:

$$\text{GTD}_1 = \sqrt{1 - \frac{r_{s,1}}{R_1}} = \sqrt{1 - \frac{1.48523 \times 10^3}{1.00000 \times 10^8}}$$

$$= \sqrt{1 - 1.48523 \times 10^{-5}} = 1 - 7.42619 \times 10^{-6}$$

$$\text{GTD}_2 = \sqrt{1 - \frac{r_{s,2}}{R_2}} = \sqrt{1 - \frac{1.48523 \times 10^0}{1.00000 \times 10^7}}$$

$$= \sqrt{1 - 1.48523 \times 10^{-7}} = 1 - 7.42616 \times 10^{-8}$$

giving the relative pace of time:

$$\frac{\text{GTD}_1}{\text{GTD}_2} = \frac{1 - 7.42619 \times 10^{-6}}{1 - 7.42616 \times 10^{-8}} = 1 - 7.35193 \times 10^{-6}$$

See Table 2, column $\rho = \rho$.

6.4 Constant Inertia ($I_1 = I_2$)

When the two systems share the same moment of inertia, $M_1 R_1^2 = M_2 R_2^2$, so $M_1/M_2 = (R_2/R_1)^2$. Substituting into k_d:

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \left(\frac{R_2}{R_1}\right)^2 \cdot \frac{R_2}{R_1} = \left(\frac{R_2}{R_1}\right)^3$$

The DeGerlia compactness ratio reduces to the cube of the radius ratio.

Example (Constant Inertia): S_1: $M_1 = 1.00000 \times 10^{32}$ kg, $R_1 = 1.00000 \times 10^7$ m. S_2: $M_2 = 1.00000 \times 10^{30}$ kg, $R_2 = 1.00000 \times 10^8$ m. Both have $I = MR^2 = 1.00000 \times 10^{46}$ kg·m^2.

DeGerlia compactness ratio:

$$k_d = \left(\frac{R_2}{R_1}\right)^3 = \left(\frac{1.00000 \times 10^8}{1.00000 \times 10^7}\right)^3 = 1.00000 \times 10^3$$

GTD calculation: Schwarzschild gravitational time dilation directly. Each system's Schwarzschild radius is $r_s = 2GM/c^2$:

$$r_{s,1} = \frac{2GM_1}{c^2} = \frac{2(6.67430 \times 10^{-11})(1.00000 \times 10^{32})}{(2.99792 \times 10^8)^2}$$

$$= 1.48523 \times 10^5 \text{ m}$$

$$r_{s,2} = \frac{2GM_2}{c^2} = \frac{2(6.67430 \times 10^{-11})(1.00000 \times 10^{30})}{(2.99792 \times 10^8)^2}$$

$$= 1.48523 \times 10^3 \text{ m}$$

GTD for each system via $\sqrt{1 - r_s/R}$:

$$\text{GTD}_1 = \sqrt{1 - \frac{r_{s,1}}{R_1}} = \sqrt{1 - \frac{1.48523 \times 10^5}{1.00000 \times 10^7}}$$

$$= \sqrt{1 - 1.48523 \times 10^{-2}} = 1 - 7.45394 \times 10^{-3}$$

$$\text{GTD}_2 = \sqrt{1 - \frac{r_{s,2}}{R_2}} = \sqrt{1 - \frac{1.48523 \times 10^3}{1.00000 \times 10^8}}$$

$$= \sqrt{1 - 1.48523 \times 10^{-5}} = 1 - 7.42619 \times 10^{-6}$$

giving the relative pace of time:

$$\frac{\text{GTD}_1}{\text{GTD}_2} = \frac{1 - 7.45394 \times 10^{-3}}{1 - 7.42619 \times 10^{-6}} = 1 - 7.44657 \times 10^{-3}$$

See Table 2, column $I = I$.

6.5 General Case (unconstrained)

When no constraint is applied between the systems, k_d stays in its full form:

$$k_d = \frac{M_1 R_2}{M_2 R_1}$$

Both mass and radius must be specified independently for each system.

Example (General Case): S_1: $M_1 = 1.00000 \times 10^{20}$ kg, $R_1 = 2.00000 \times 10^{-7}$ m. S_2: $M_2 = 1.00000 \times 10^{30}$ kg, $R_2 = 1.00000 \times 10^7$ m.

DeGerlia compactness ratio:

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \frac{(1.00000 \times 10^{20})(1.00000 \times 10^7)}{(1.00000 \times 10^{30})(2.00000 \times 10^{-7})} = 5.00000 \times 10^3$$

GTD calculation: Schwarzschild gravitational time dilation directly. Each system's Schwarzschild radius is $r_s =$

$2GM/c^2$:

$$r_{s,1} = \frac{2GM_1}{c^2} = \frac{2(6.67430\times10^{-11})(1.00000\times10^{20})}{(2.99792\times10^{8})^2}$$

$$= 1.48523\times10^{-7}\,\mathrm{m}$$

$$r_{s,2} = \frac{2GM_2}{c^2} = \frac{2(6.67430\times10^{-11})(1.00000\times10^{30})}{(2.99792\times10^{8})^2}$$

$$= 1.48523\times10^{3}\,\mathrm{m}$$

GTD for each system via $\sqrt{1-r_s/R}$:

$$\mathrm{GTD}_1 = \sqrt{1-\frac{r_{s,1}}{R_1}} = \sqrt{1-\frac{1.48523\times10^{-7}}{2.00000\times10^{-7}}}$$

$$= \sqrt{1-7.42617\times10^{-1}} = 1-4.92670\times10^{-1}$$

$$\mathrm{GTD}_2 = \sqrt{1-\frac{r_{s,2}}{R_2}} = \sqrt{1-\frac{1.48523\times10^{3}}{1.00000\times10^{7}}}$$

$$= \sqrt{1-1.48523\times10^{-4}} = 1-7.42644\times10^{-5}$$

giving the relative pace of time:

$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = \frac{1-4.92670\times10^{-1}}{1-7.42644\times10^{-5}} = 1-4.92632\times10^{-1}$$

See Table 2, column General.

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Conflicts of Interest

The author declares no competing interests.

Data Availability

The complete source materials for this paper, including all LaTeX source files, derivation documents, and revision history, are publicly available at: `https://github.com/denverdata/academic_InertialRelativity-deriving-gr-sr-from-ste`.

References

[1] T. D. DeGerlia, *Introducing Inertial Density and Characterizing the Schwarzschild Condition as a Constant Threshold of Mass over Radius (m/r)*, DEC, 2025, DOI: 10.55277/researchhub.0e0cgoor.

[2] A. Einstein, *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften (Berlin)*, 1915, 844–847.

[3] K. Schwarzschild, *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften (Berlin)*, 1916, 189–196.

[4] A. Einstein, *Annalen der Physik*, 1905, **17**, 891–921.

[5] E. Tiesinga, P. J. Mohr, D. B. Newell and B. N. Taylor, *Reviews of Modern Physics*, 2021, **93**, 025010.

[6] T. D. DeGerlia, *The Universe of Light - Exploring our Universe through the Lens of the Law of Space-Time Equivalence*, DEC, 2025, DOI: 10.55277/ResearchHub.pi9yjshm.

[7] H. Goldstein, C. Poole and J. Safko, *Classical Mechanics*, Addison-Wesley, 3rd edn, 2002, pp. 70–71.

Appendix A Pair Comparison Tables

Tables 1 and 2 present the complete system properties and derived ratios for six system pairs spanning the constraint cases from the derivations in Section 6: static mass (M=M), static radius (R=R), constant density ($\rho = \rho$), constant inertia (I=I), an unconstrained general case, and an STE example. The three algebraic forms of k_d in Equation 10 produce identical values in every case, alongside the gravitational time-dilation ratio computed independently from the standard Schwarzschild form.

Table 1 System properties for each pair comparison case.

	Case 1 ($M = M$)	**Case 2** ($R = R$)	**Case 3** ($\rho = \rho$)	**Case 4** ($I = I$)	**General**	**STE Example**
m_1 (kg)	2.00000e+30	1.00000e+30	1.00000e+30	1.00000e+32	1.00000e+20	8.00000e+0
m_2 (kg)	2.00000e+30	1.00000e+24	1.00000e+27	1.00000e+30	1.00000e+30	1.00000e+0
r_1 (m)	3.00000e+3	1.00000e+8	1.00000e+8	1.00000e+7	2.00000e-7	2.00000e+0
r_2 (m)	7.00000e+8	1.00000e+8	1.00000e+7	1.00000e+8	1.00000e+7	1.00000e+0
ρ_1 (kg/m^3)	1.76839e+19	2.38732e+5	2.38732e+5	2.38732e+10	2.98416e+39	2.38732e-1
ρ_2 (kg/m^3)	1.39203e+3	2.38732e-1	2.38732e+5	2.38732e+5	2.38732e+8	2.38732e-1
I_1 (kg·m^2)	7.20000e+36	4.00000e+45	4.00000e+45	4.00000e+45	1.60000e+6	1.28000e+1
I_2 (kg·m^2)	3.92000e+47	4.00000e+39	4.00000e+40	4.00000e+45	4.00000e+43	4.00000e-1
D_1 (kg/m)	6.66667e+26	1.00000e+22	1.00000e+22	1.00000e+25	5.00000e+26	4.00000e+0
D_2 (kg/m)	2.85714e+21	1.00000e+16	1.00000e+20	1.00000e+22	1.00000e+23	1.00000e+0
D_{1_norm}	9.90155e-1	1.48523e-5	1.48523e-5	1.48523e-2	7.42617e-1	5.94093e-27
D_{2_norm}	4.24352e-6	1.48523e-11	1.48523e-7	1.48523e-5	1.48523e-4	1.48523e-27
r_{s_1} (m)	2.97046e+3	1.48523e+3	1.48523e+3	1.48523e+5	1.48523e-7	1.18819e-26
r_{s_2} (m)	2.97046e+3	1.48523e-3	1.48523e+0	1.48523e+3	1.48523e+3	1.48523e-27
gtd_1	1-9.00777e-1	1-7.42619e-6	1-7.42619e-6	1-7.45394e-3	1-4.92670e-1	1-2.97046e-27
gtd_2	1-2.12176e-6	1-7.42616e-12	1-7.42616e-8	1-7.42619e-6	1-7.42644e-5	1-7.42616e-28

Table 2 Derived ratios across constraint cases.

	Case 1 ($M = M$)	**Case 2** ($R = R$)	**Case 3** ($\rho = \rho$)	**Case 4** ($I = I$)	**General**	**STE Example**
$\sqrt{D_1/D_2}$	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$\sqrt{(m_1 r_2)/(m_2 r_1)}$	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$\sqrt{(I_1/I_2)(r_2/r_1)^3}$	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$(I_1/I_2)^{1/5}(\rho_1/\rho_2)^{3/10}$	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$(m_1 r_2)/(m_2 r_1)$	2.33333e+5	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
D_1/D_2	2.33333e+5	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
$(I_1/I_2)(r_2/r_1)^3$	2.33333e+5	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
gtd_1/gtd_2	1-9.00776e-1	1-7.42618e-6	1-7.35193e-6	1-7.44657e-3	1-4.92632e-1	1-2.22785e-27
I_1/I_2	1.83673e-11	1.00000e+6	1.00000e+5	1.00000e+0	4.00000e-38	3.20000e+1
$(I_1/I_2)^{1/2}$	4.28571e-6	1.00000e+3	3.16228e+2	1.00000e+0	2.00000e-19	5.65685e+0
$(\rho_1/\rho_2)^{1/5}(r_1/r_2)$	7.12540e-3	1.58489e+1	1.00000e+1	1.00000e+0	3.31445e-8	2.00000e+0
$(I_1/I_2)^{1/5}$	7.12540e-3	1.58489e+1	1.00000e+1	1.00000e+0	3.31445e-8	2.00000e+0
m_1/m_2	1.00000e+0	1.00000e+6	1.00000e+3	1.00000e+2	1.00000e-10	8.00000e+0
$\sqrt{m_1/m_2}$	1.00000e+0	1.00000e+3	3.16228e+1	1.00000e+1	1.00000e-5	2.82843e+0
r_1/r_2	4.28571e-6	1.00000e+0	1.00000e+1	1.00000e-1	2.00000e-14	2.00000e+0
$\sqrt{r_1/r_2}$	2.07020e-3	1.00000e+0	3.16228e+0	3.16228e-1	1.41421e-7	1.41421e+0